

Arnav Kakkar

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Education

Thapar Institute of Engineering and Technology, Patiala	Sep 2023 – Jun 2027
B.E. Electronics Engineering	
The Vivekanand School, New Delhi	Apr 2021 – Jun 2023
CBSE Senior Secondary	

Experience

Software Development Engineer (SDE) Intern	Feb 2026 – Mar 2026
Bluestock Fintech — Remote	

- Completed technical assessment with a score of 100/100 and contributed to React, Node.js, and Firebase-based applications with REST API integration, improving responsiveness and usability.
- Developed UI components using React and Tailwind CSS while supporting feature implementation and workflow optimization in a Git-based environment, ensuring clean code and consistent interface design.

Projects

RAG-Based AI Teaching Assistant	Jan 2026 – Feb 2026
Python, LangChain, FAISS, OpenAI API, NLP	

- Built a Retrieval-Augmented Generation (RAG) system using LangChain and FAISS to query 500+ documents, enabling fast and context-aware responses across 50+ queries per session.
- Developed an embedding-based semantic search pipeline, increasing answer relevance by 30% and reducing incorrect outputs by 25% through improved context retrieval.
- Optimized retrieval and response workflow, reducing query latency by 35% while supporting concurrent users with stable and efficient performance.

Real-Time Anomaly Detection System	Apr 2026 – May 2026
Python, Scikit-learn, Streamlit, Plotly	

- Built a real-time anomaly detection system using Isolation Forest on 1000+ streaming data points with continuous tracking, monitoring, and detection of unusual patterns in data.
- Created 5+ time-series features and combined ML with rule-based logic, improving anomaly detection accuracy by 20% across different test cases and varying data conditions.
- Developed a Streamlit dashboard to show 100+ live data points and generate 3–4 clear explanations per anomaly, helping users better understand patterns and take decisions.

Leadership

Executive Core Member	Feb 2024 – Jul 2025
Frosh (under the aegis of Dean of Student Affairs)	Thapar Institute of Engineering and Technology, Patiala

- Led **79 mentors** as part of **20-member core team** handling **20,000+ applicants**.
- Coordinated **30-day Fresher's Fest** with **3,200+ participants**.
- Headed Media Department for event coverage and coordination.

Executive Committee Member — Saturnalia Tech Fest	Sep 2025 – Nov 2025
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- Led media operations for 400+ member team.
- Increased inter-college participation through outreach.
- Executed events with **3000+ attendees**.

Certifications

The Ultimate Job Ready Data Science Course	Nov 2025 - Mar 2026
codewithharry.com	

Decode DSA with C++	Mar 2026 - Jul 2026
PW Skills	

Skills

Languages: Python, JavaScript, C, SQL

Frameworks/Tools: React.js, Firebase, Pandas, NumPy, Matplotlib, Git, GitHub, Vercel, Jupyter Notebook

Core Skills: Data Structures, Algorithms, REST APIs, Authentication Systems, Full-Stack Development, Data Analysis, Debugging

AI/ML: RAG, LLM Integration, Semantic Search, Data Preprocessing, EDA